

For the purposes of developing tests, it will be useful to re-cast this null hypothesis in terms of moments involving an unrestricted nuisance parameter $\tilde{\tau}$ of dimension $\bar{T} - 1$. By applying a change of basis to the matrix AL_{post} , we can re-write the expression $AL_{post}\tau_{post}$ as $\tilde{A}(\theta, \tilde{\tau})'$ for $\tilde{\tau} \in \mathbb{R}^{\bar{T}-1}$.¹³ The null H_0 is then equivalent to

$$H_0 : \exists \tilde{\tau} \in \mathbb{R}^{\bar{T}-1} \text{ s.t. } \mathbb{E}[\tilde{Y}_n(\bar{\theta}) - \tilde{X}\tilde{\tau}] \leq 0, \quad (13)$$

where $\tilde{Y}_n(\bar{\theta}) = Y_n - \tilde{A}_{(\cdot,1)}\bar{\theta}$ and $\tilde{X} = \tilde{A}_{(\cdot,-1)}$.¹⁴ Since $\tilde{Y}_n(\bar{\theta})$ is normally distributed with covariance matrix $\tilde{\Sigma}_n = A\Sigma_n A'$ under the finite-sample normal model, testing $H_0 : \theta = \bar{\theta}, \delta \in \Delta$ is equivalent to testing a system of moment inequalities with linear nuisance parameters.

The testing problem (13) is a special case of the problem studied in ARP, which focuses on testing null hypotheses of the form $H_0 : \exists \tau \text{ s.t. } \mathbb{E}[Y(\theta) - X\tau | X] \leq 0$ (a.s.). Our setting is a special case of this framework in which: i) the variable X takes the degenerate distribution $X = \tilde{X}$, and ii) $Y(\theta) = \tilde{Y}(\theta)$ is linear in θ . This additional structure will play an important role in the development of our asymptotic power results below.

3.2. Constructing conditional and hybrid confidence sets

We consider tests for the system of moment inequalities described above using the conditional and hybrid methods proposed by ARP. This is for both computational and efficiency reasons. From the computational perspective, a practical challenge to testing the hypothesis (13) in our setting is that the dimension of the nuisance parameter $\tilde{\tau}$ is $\bar{T} - 1$, and thus will be large if there are many post-treatment periods. For example, 5 of the 12 recent event-study papers reviewed in Roth (2022) have $\bar{T} > 10$. This renders many moment inequality methods, especially those which rely on test inversion over a grid for the full parameter vector, computationally infeasible. To tractably deal with the nuisance parameter, we consider tests based on the conditional and hybrid approaches of ARP, which directly exploit the linear structure of the hypothesis (13) to deliver computationally tractable tests even when the number of post-treatment periods $\bar{T}$ is large.¹⁵ From the perspective of power, we will show that the tests proposed by ARP have (near-)optimal local asymptotic power in our setting when an LICQ condition is satisfied.

We briefly sketch the construction of the conditional testing approach in our setting, and refer the reader to ARP for full details. These tests are implemented in the HonestDiD R and Stata packages that accompany the paper.

13. Let Γ be a square matrix with the vector l' in the first row and remaining rows chosen so that Γ has full rank. Define $\tilde{A} := AL_{post}\Gamma^{-1}$. Then $AL_{post}\tau_{post} = \tilde{A}\Gamma\tau_{post} = \tilde{A}\begin{pmatrix} \theta \\ \underbrace{\Gamma_{(-1,\cdot)}\tau_{post}}_{:=\tilde{\tau}} \end{pmatrix}$. If $\bar{T} = 1$, then $\tilde{\tau}$ is 0-dimensional and should be interpreted as 0.

14. We use the notation $V_{(\cdot,1)}$ to denote the first column of a matrix V , and $V_{(\cdot,-1)}$ to denote the matrix containing all but the first column of V .

15. Other moment inequality methods have been proposed for subvector inference, but typically do not exploit the linear structure of our setting—see, e.g. Romano and Shaikh (2008), Chernozhukov *et al.* (2015), Bugni *et al.* (2017) and Chen *et al.* (2018) and Kaido *et al.* (2019). Cho and Russell (2019), Gafarov (2019), and Flynn (2019) also provide methods for subvector inference with linear moment inequalities, but in contrast to our approach require a linear independence constraint qualification (LICQ) assumption for size control. More recently, Cox and Shi (2022) introduced new tests for the linear moment inequality setting in ARP; see Section 3.5 below for further discussion.

3.2.1. Conditional confidence sets. Suppose we wish to test (13) for some fixed $\bar{\theta}$. The conditional testing approach considers tests based on the profiled test statistic

$$\hat{\eta} := \min_{\eta, \tilde{\tau}} \eta \text{ s.t. } \tilde{Y}_n(\bar{\theta}) - \tilde{X}\tilde{\tau} \leq \tilde{\sigma}_n \cdot \eta, \quad (14)$$

where $\tilde{\sigma}_n = \sqrt{\text{diag}(\tilde{\Sigma}_n)}$. This linear program selects the value of the nuisance parameters $\tilde{\tau} \in \mathbb{R}^{\tilde{T}-1}$ that minimizes the maximum studentized moment. Duality results from linear programming (e.g. Schrijver (1986), Section 7.4) imply that the value $\hat{\eta}$ obtained from the primal program (14) equals the optimal value of the dual program¹⁶

$$\hat{\eta} = \max_{\gamma} \gamma' \tilde{Y}_n(\bar{\theta}) \text{ s.t. } \gamma' \tilde{X} = 0, \gamma' \tilde{\sigma}_n = 1, \quad \gamma \geq 0. \quad (15)$$

If a vector γ_* is optimal in the dual problem above, then it is a vector of Lagrange multipliers for the primal problem. Standard results in linear programming imply that the optimum is always obtained at one of the finite set of vertices, $V(\Sigma_n)$ (also known as the set of basic feasible solutions). We denote by $\hat{V}_n \subset V(\Sigma_n)$ the set of optimal vertices of the dual program.¹⁷

To construct critical values, ARP use the fact that the distribution of $\hat{\eta}$ has a truncated normal distribution conditional on the event that γ_* is optimal in the dual problem. Specifically,

$$\hat{\eta} | \{\gamma_* \in \hat{V}_n, S_n = s\} \sim \xi | \xi \in [v^{lo}, v^{up}],$$

where $\xi \sim \mathcal{N}(\gamma'_* \tilde{\mu}(\bar{\theta}), \gamma'_* \tilde{\Sigma}_n \gamma_*)$, $\tilde{\mu}(\bar{\theta}) = \mathbb{E}[\tilde{Y}_n(\bar{\theta})]$, $S_n = (I - (\tilde{\Sigma}_n \gamma_*/\gamma'_* \tilde{\Sigma}_n \gamma_*) \gamma'_*) \tilde{Y}_n(\bar{\theta})$, and v^{lo}, v^{up} are known functions of $\tilde{\Sigma}_n, s, \gamma_*$ (see Lemma 1 in ARP).¹⁸ Intuitively, the distribution of $\hat{\eta}$ depends on the vector $\tilde{\mu}(\bar{\theta})$, and so to eliminate the dependence on the components of $\tilde{\mu}(\bar{\theta})$ other than $\gamma'_* \tilde{\mu}(\bar{\theta})$, we condition on S_n , which is a sufficient statistic for the components of $\tilde{\mu}(\bar{\theta})$ that are orthogonal to $\gamma'_* \tilde{\mu}(\bar{\theta})$.

ARP show that all quantiles of the conditional distribution of $\hat{\eta}$ in the previous display are increasing in $\gamma'_* \tilde{\mu}(\bar{\theta})$. Moreover, the null hypothesis (13) implies $\gamma'_* \tilde{\mu}(\bar{\theta}) \leq 0$. To see why this is the case, note that the definition of the dual problem (15) implies that $\gamma_* \geq 0$ and $\gamma'_* \tilde{X} = 0$, whereas the null hypothesis implies that there exists $\tilde{\tau}$ such that $\tilde{\mu}(\bar{\theta}) - \tilde{X}\tilde{\tau} \leq 0$. It follows that $\gamma'_* \tilde{\mu}(\bar{\theta}) = \gamma'_* (\tilde{\mu}(\bar{\theta}) - \tilde{X}\tilde{\tau}) \leq 0$ under the null. The ARP conditional test therefore uses the critical value $\max\{0, c_{C,\alpha}\}$, where $c_{C,\alpha}$ is the $1 - \alpha$ quantile of the truncated normal distribution $\xi | \xi \in [v^{lo}, v^{up}]$ under the worst-case assumption that $\gamma'_* \tilde{\mu}(\bar{\theta}) = 0$.¹⁹ We denote by

16. Technically, the duality results require that $\hat{\eta}$ be finite. However, one can show that $\hat{\eta}$ is finite with probability 1, unless the span of $\tilde{X}$ contains a vector with all negative entries, in which case the identified set for θ is the real line. We therefore trivially define our test never to reject if $\hat{\eta} = -\infty$.

17. In general, there may not be a unique solution to the dual program. ARP show that in the context of the finite sample normal, conditional on any one vertex of the dual program's feasible set being optimal, every other vertex is optimal with either probability 0 or 1. In the finite sample normal model, it thus suffices to condition on the event that a vector $\gamma_* \in \hat{V}$. Our conditions for asymptotic validity of the conditional test below, however, ensure that the optimal vertex will be unique w.p.a. 1.

18. The cut-offs v^{lo} and v^{up} are the maximum and minimum of the set $\{x : x = \max_{\gamma \in F_n} \gamma'(s + (\tilde{\Sigma}_n \gamma_*/\gamma'_* \tilde{\Sigma}_n \gamma_*)x)\}$ when $\gamma'_* \tilde{\Sigma}_n \gamma_* \neq 0$, where F_n is the feasible set of the dual program (15). When $\gamma'_* \tilde{\Sigma}_n \gamma_* = 0$, we define $v^{lo} = -\infty$ and $v^{up} = \infty$, so the conditional test rejects if and only if $\hat{\eta} > 0$.

19. As noted in ARP, the truncation at 0 is not necessary for the conditional test to control size in the finite sample normal model, but it simplifies asymptotic arguments. It also prevents the test from rejecting when all moments are satisfied in sample.

$\psi_\alpha^C(\hat{\beta}_n, A, d, \bar{\theta}, \Sigma_n)$ an indicator for whether the conditional test rejects the null that $\theta = \bar{\theta}$ for $\Delta = \{\delta : A\delta \leq d\}$.

We can then form a confidence set for θ by test inversion, $\mathcal{C}_{\alpha,n}^C(\hat{\beta}_n, \Sigma_n) := \{\bar{\theta} : \psi_\alpha^C(\hat{\beta}_n, A, d, \bar{\theta}, \Sigma_n) = 0\}$. The construction of the conditional test implies that $\mathbb{E}_{\hat{\beta}_n \sim \mathcal{N}(\delta + \tau, \Sigma_n)} [\psi_\alpha^C(\hat{\beta}_n, A, d, \tau_{post}, \Sigma_n)] \leq \alpha$ for any $\delta \in \Delta$. It therefore follows that $\mathcal{C}_{\alpha,n}^C(\hat{\beta}_n, \Sigma_n)$ satisfies the finite-sample coverage requirement (10). In Section 3.3 below, we show that coverage in the normal model translates to uniform asymptotic coverage over a large class of DGPs.

Example 4. An instructive example is when $\bar{T} = 1$ (so that there are no nuisance parameters), and $\tilde{\Sigma}_n = I$. Then $\hat{\eta} = \max_j \tilde{Y}_{n,j}$ is the maximum component of $\tilde{Y}_n$, $v^{lo} = \max_{j \neq \hat{j}} \tilde{Y}_{n,j}$ is the second-largest element of $\tilde{Y}_n$ (where $\hat{j}$ denotes the index of the max), and $v^{up} = \infty$. Thus, the conditional test rejects when $\hat{\eta}$ exceeds the $1 - \alpha$ quantile of the standard normal distribution truncated to $[v^{lo}, \infty)$. Intuitively, this means that the conditional test will tend to reject when the maximum sample moment is far enough away from the second-largest sample moment. Two special cases are worth special consideration. First, consider the case where in population one moment is violated and the remaining moments are very slack, e.g. $\tilde{\mu}_1 > 0$ while $\tilde{\mu}_j \ll 0$ for $j \neq 1$. Then with high probability $\hat{\eta}$ will equal $\tilde{Y}_{n,1}$ and v^{lo} will be very negative. Thus, the conditional test will behave similarly to a one-sided t -test using $\tilde{Y}_{n,1}$, which can be shown to be the most powerful test in the finite-sample normal model in this example. On the other hand, if $\mu_1 \approx \mu_2 > 0$, then the maximum and second-largest sample moments (i.e. $\hat{\eta}$ and v^{lo}) will be close together with high probability, so the conditional test may not reject with substantial probability even if both μ_1 and μ_2 are large, and thus the conditional test may have poor power. To improve power in these settings where the binding and non-binding moments are close together (relative to sampling variation), ARP introduce a “hybrid” test, which we describe next.

3.2.2. Hybrid confidence sets. ARP propose a “hybrid” test that combines the conditioning approach above with a test based on the “least-favourable” assumption that $\tilde{\mu}(\bar{\theta}) = 0$. In particular, ARP show that the distribution of $\hat{\eta}$ under the null is bounded above (in the sense of first-order stochastic dominance) by the distribution of $\hat{\eta}$ when $\tilde{\mu}(\bar{\theta}) = 0$ (see Section 3.2 of ARP). One can therefore construct a size- κ least-favourable (LF) test in the finite-sample normal model that rejects whenever $\hat{\eta}$ exceeds the $1 - \kappa$ quantile of $\max_{\gamma \in V(\Sigma)} \gamma' \tilde{\zeta}$, where $\tilde{\zeta} \sim \mathcal{N}(0, \tilde{\Sigma}_n)$. This critical value, which we will denote by $c_{LF,\kappa}$, can easily be calculated by simulation. For $0 < \kappa < \alpha$, the ARP conditional-LF hybrid test is defined to reject if a first-stage, size- κ LF test rejects. If this first-stage test does not reject, then in the second stage the hybrid test conducts a modified version of the size- $((\alpha - \kappa)/(1 - \kappa))$ conditional test that also conditions on the event that the first-stage LF test did not reject. In particular, by similar logic as for the conditional test, we have that

$$\hat{\eta} | \{\gamma_* \in \hat{V}_n, S_n = s, \hat{\eta} \leq c_{LF,\kappa}\} \sim \tilde{\zeta} | \tilde{\zeta} \in [v^{lo}, v_H^{up}],$$

where $v_H^{up} = \min\{v^{lo}, c_{LF,\kappa}\}$ (see Section 3.4 of ARP). The second-stage of the hybrid test rejects if $\hat{\eta}$ exceeds the critical value for the size- $((\alpha - \kappa)/(1 - \kappa))$ conditional test that uses v_H^{up} instead of v^{up} . We will denote by $\psi_{\kappa,\alpha}^{C-LF}(\hat{\beta}_n, A, d, \bar{\theta}, \Sigma_n)$ an indicator for whether the hybrid test rejects at a particular value $\bar{\theta}$, and denote by $\mathcal{C}_{\kappa,\alpha,n}^{C-LF}(\hat{\beta}_n, \Sigma_n)$ the confidence set that collects the values of $\bar{\theta}$ for which the hybrid test does not reject. As with the conditional test, by construction the hybrid confidence set satisfies that coverage criterion (10) in the finite-sample normal model. In our implementation below, we use $\kappa = \alpha/10$, following ARP.

3.3. Uniform asymptotic size control

We now provide conditions under which size control in the finite sample normal model translates to uniform asymptotic size control over a large class of data-generating processes $\mathcal{P}$ under which $\hat{\beta}_n$ is asymptotically normally distributed and Σ_n is replaced with a consistent estimator $\hat{\Sigma}_n$. In particular, we provide sufficient conditions on $\hat{\beta}_n$, $\hat{\Sigma}_n$, and Δ such that the higher-level conditions for size control in ARP are satisfied (Proposition 2 in ARP).

Throughout this section, we fix $\Delta = \{A\delta \leq d\}$ for some A with all non-zero rows, and assume that Δ is non-empty. We consider a class of data-generating processes, indexed by $P \in \mathcal{P}$, under which $\sqrt{n}(\hat{\beta}_n - \beta_P)$ is asymptotically normal, where β_P satisfies the causal decomposition in (3), i.e. $\beta_P = \delta_P + L_{post}\tau_{P,post}$ for $\delta_P \in \Delta$ and $\tau_{P,post} \in \mathbb{R}^T$. The parameter of interest is $\theta_P := l'\tau_{P,post}$, for some fixed $l \neq 0$. Our first assumption imposes uniform asymptotic normality of $\hat{\beta}_n$.

Assumption 2. Let BL_1 denote the set of Lipschitz functions which are bounded by 1 in absolute value and have Lipschitz constant bounded by 1. We assume

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \sup_{f \in BL_1} \left| \mathbb{E}_P \left[f(\sqrt{n}(\hat{\beta}_n - \beta_P)) \right] - \mathbb{E} \left[f(\xi_P) \right] \right| = 0,$$

where $\xi_P \sim \mathcal{N}(0, \Sigma_P)$, and $\beta_P = \delta_P + L_{post}\tau_{P,post}$ for $\delta_P \in \Delta$ and $\tau_{P,post} \in \mathbb{R}^T$.

Convergence in distribution is equivalent to convergence in bounded Lipschitz metric (see Theorem 1.12.4 in van der Vaart and Wellner, 1996), so Assumption 2 formalizes the notion of uniform convergence in distribution of $\sqrt{n}(\hat{\beta}_n - \beta_P)$ to a $\mathcal{N}(0, \Sigma_P)$ variable under P .

Our next two assumptions require that the eigenvalues of the asymptotic variance of $\hat{\beta}_n$ be bounded above and away from zero, and that there exists a uniformly consistent estimator for the variance of $\hat{\beta}_n$.

Assumption 3. Let $\mathbf{S}$ denote the set of matrices with eigenvalues bounded below by $\underline{\lambda} > 0$ and above by $\bar{\lambda} \geq \underline{\lambda}$. For all $P \in \mathcal{P}$, $\Sigma_P \in \mathbf{S}$.

Assumption 4. We have an estimator $\hat{\Sigma}_n$ that is uniformly consistent for Σ_P ,

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \mathbb{P}_P \left(\|\hat{\Sigma}_n - \Sigma_P\| > \epsilon \right) = 0,$$

for all $\epsilon > 0$.

Finally, we impose some regularity conditions on the matrix A .

Assumption 5. At least one of the following holds.

- (A) For $k_1 + k_2 = \dim(\delta)$, the matrix A can be written as TQ , where Q has full row-rank and $T = \begin{pmatrix} I_{k_1} & 0 \\ -I_{k_1} & 0 \\ 0 & I_{k_2} \end{pmatrix}$. (We allow for the case where one of k_1 or k_2 is 0, in which case the zero-dimensional blocks can be ignored).
- (B) Let $\bar{\gamma}_1, \dots, \bar{\gamma}_K$ be the elements of $V(I)$. Then for all k , either $\bar{\gamma}_k' A = 0$ or $\inf_{a \geq 0} \inf_{j \neq k} \|(\bar{\gamma}_k - a\bar{\gamma}_j)' A\| > 0$.

Part (A) of Assumption 5 imposes that the only source of degeneracy in the rows of A is matching inequalities of opposite signs. This is the case for many restrictions of interest, such as $\Delta^{\text{SD}}(M)$ and the polyhedra that form $\Delta^{\text{RM}}(\bar{M})$. Part (B) provides an alternative, higher-level condition that ensures that for distinct vertices $\bar{\gamma}_k, \bar{\gamma}_j$, the random variables $\bar{\gamma}_j' \tilde{Y}_n$ and $\bar{\gamma}_k' \tilde{Y}_n$